

Satyam Model 2

Technical Report

Bacterial vs Fungal keratitis from slit-lamp photographs — development, evaluation, confound analysis, segmentation & interpretability. Complete record to date, before expansion to more data and additional organisms.

0.815

locked internal test AUC (n=131)

0.839

pooled external AUC (n=164, 2 cohorts)

0/22

dangerous errors, blind head-to-head

Executive summary

Satyam Model 2 classifies culture-proven bacterial vs fungal keratitis from a single slit-lamp image. It is a ground-up rebuild of LVPEI's incumbent CornealAI Model 2, whose headline numbers are not trustworthy. On a patient-disjoint locked test it reaches **AUC 0.815**, and on 164 genuinely independent external cases **0.839** — the first uncontaminated measurement of this task. It reads pathology (not the camera), localises to the lesion, abstains when unsure, and shows the clinician what it saw.

The one-line result

On an 81%-fungal blind set it matched the incumbent on accuracy but made the sight-threatening error — a fungal ulcer called bacterial — **zero times versus the incumbent's eight**.

The incumbent problem

CornealAI Model 2's quoted **AUC 0.949** is computed over all 686 images, 548 of which it trained on. Its best honest figure (0.862 / 77.5% on a 138-image split) is also contaminated: the bag builder inserts a lesion tile when the label is bacterial and a hypopyon tile when it is fungal — **in training and validation** — while inference does not, so the distributions differ. No independent test set, no patient grouping. **Neither number is comparable to a clean held-out figure**. That gap is what this project set out to close.

Design rules (held throughout)

- **Patient-grouped, repeated nested CV**. One split has $sd \approx 0.015$ (± 0.03 noise) — enough to move a verdict a category. Every AUC is $mean \pm sd$ over ≥ 8 –10 fold assignments.
- **Nothing about the label touches preprocessing or tiling** — the exact defect that invalidated the incumbent.
- **Aspect ratio preserved** (letterbox, never tuple-resize — that distorts the limbus by a device-dependent factor).
- **Metrics at deployment prevalence**, not the curated 1:1, and always with abstention.
- **Ties are called ties**. Several promising ideas tied the baseline and were not adopted.

Cohort

Property	Value
Images / patients	682 / 617 (1.11 img/patient)
Classes	335 bacterial / 347 fungal
Label source	culture-proven (organism filed by culture result)
Acquisition	Canon EOS 7D Mk II, ~5472×3648; 3 resolutions
Split	131 locked test · 551 dev (5-fold), patient-disjoint, assigned once
Integrity	0 duplicate hashes · 0 unreadable · 0 patients spanning splits

De-identification

Names/MRNs were in every filename; EXIF held timestamps, camera identity and thumbnails. All removed by **lossless JPEG marker surgery** (pixel-identical verified — re-encoding would have degraded the fine margin texture the model depends on). Verification: **0 name/MRN tokens** survive outside the access-controlled crosswalk.

Confound audit — is the class readable without pathology?

Predictor	AUC	Verdict
Metadata (date, centre, resolution, file size)	0.577	clean
Global image statistics (22 features)	0.543	clean
Background tiles (sclera / lid / lashes)	0.582	clean
Worst camera-like perturbation (bright/gamma/hue/JPEG/blur)	−0.005 vs clean	robust

Reads pathology, not the camera

Nothing outside the cornea predicts the class ($0.58 \approx \text{chance}$), and the decision barely moves under camera-like changes. Whatever the model has learned, it learned from the cornea.

Evaluation protocol

Repeated nested cross-validation: an inner loop selects hyper-parameters on training folds only, an outer loop reports, over ≥ 8 fold assignments, quoted mean $\pm$ sd. The locked 131-image test set was untouched from the first day until the single final evaluation. This discipline is why the numbers here can be trusted where the incumbent's cannot.

Where does the signal live?

A frozen DINOv2 probe (linear head only) established that whole-eye representations plateau at ~0.745 regardless of resolution or framing — global pooling averages the fine texture away. The lever turned out to be **tile field of view**. The scale curve is unimodal and peaks at 3.67 mm — which coincides with the 3.44 mm span of the feathery-margin sign in expert annotations, derived independently.

Tile field of view	AUC
0.92 mm	0.745
1.83 mm	0.759
3.67 mm — peak	0.791 ± 0.009
5.50 mm	0.775
7.33 mm	0.722
11.7 mm (whole eye)	0.747

What did NOT help (recorded, because it shaped the design)

Idea	Result	Decision
Gated attention pooling	0.689	mean pooling wins (attention overfits at n=682)
Full pixel fidelity (896 input)	-0.009	field of view matters, not fidelity
Multi-scale bags	0.784	below 3.67 mm alone
Lesion-guided tiling	0.797 (±0.025)	tie — kept whole-cornea (simpler)
Biomarker fusion	0.800→0.799	tie — biomarkers for display only

Two real gains only: tile scale (+0.046) and 15-model ensembling (+0.024). Everything else tied or hurt — which is itself the finding that the ceiling is the data, not the architecture.

Regional ablation

The inferior third of the cornea alone — where the hypopyon and infiltrate sit — scores 0.786, ≈ the whole cornea's 0.791. The signal is concentrated in the disease region, exactly as clinical reasoning predicts.

Configuration

Native image → **limbus segmentation** (UNet++, validated 6/6 on anatomy) → **896 px native tiles (3.67 mm)** fed at 448 → **frozen DINOv2 ViT-S/14** → **mean-pooled MIL** (~75k trainable params) → **15-model ensemble** → **temperature calibration**. Frozen backbone is why it holds on 682 images.

Performance

0.806
dev out-of-fold AUC (551)

0.815
locked test AUC (131) [0.74–0.89]

0.839
pooled external (164)

0.998
calibration temperature

Three independent estimates converge at ~0.81–0.84. Test ≥ dev, so no overfitting.

External validation — two independent cohorts

Cohort	n (bac/fun)	AUC [95% CI]	In-scope	Overlap
External 1	65 (24/41)	0.905 [0.82–0.97]	65 of 100	4 (checked)
External 2	99 (35/64)	0.792 [0.70–0.88]	99 of 100	0
Pooled	164	0.839 [0.78–0.89]	—	—

Operating points (calibrated on external data)

Mode	Coverage	Accuracy	Fungal recall	Misroute
Cautious (default, abstains)	78%	81%	83%	2.9%
Fungal-safety (t=0.25)	100%	73%	97%	2.9%
Balanced (removed)	100%	77%	89%	11.4%

Leave-one-cohort-out transfer (the only unbiased operating-point figure): 76% accuracy on covered cases. Balanced was removed — it made the dangerous error 3.5× more often.

5 · Segmentation, interpretability & comparison

Lesion segmentation (from DataM2 annotations)

Two focused UNet++ models trained on LabelMe polygons — a lesion model (Infection set, 344 imgs) and a fungal-sign model (110 imgs).

Structure	Val Dice	Use
Infiltrate	0.88	anchor — overlay + biomarkers
Hypopyon	0.86	anchor — detected in 70% of cohort
Cellularity	0.57	excluded — confound (below)
Feathery margin	0.52	fungal sign detector (noisy)

Cellularity is a confound — excluded with evidence

Flagged clinically (subjective annotation; appears during healing → tracks treatment stage, not organism). Confirmed: 0.51 alone, and -0.002 when fused. It is kept out of both the classifier and the display overlay.

Interpretability, not accuracy

Biomarker fusion gave no accuracy gain (0.800→0.799) — the deep model already encodes what the geometric signs measure. So segmentation serves **interpretability**: the app outlines the detected infiltrate and hypopyon (outlines, so tissue stays visible) and shows clinical-sign chips, explicitly marked as review-only. The clinician can check the model against what they see.

Blind head-to-head vs incumbent (27 images, 22 fungal / 5 bacterial)

Metric	Incumbent M2	Satyam M2
Accuracy (forced)	66.7%	66.7%
Accuracy (when it commits)	66.7%	90.0%
Fungal called bacterial (dangerous)	8 of 22	0 of 22
Behaviour when unsure	guesses	refers to culture

Honest limitations

- **Bacterial is the binding constraint.** ~204 bacterial patients in the whole source; bacterial recall is weak in every cohort (73% / 62% / 51%). This is a data problem, not an architecture one — no modelling change fixed it.
- **Scope: binary only.** The model cannot say normal / viral / scar. In a real review cohort 35% of images were out of scope and 27/35 got confident calls. **Must be gated behind an infection detector before deployment.**
- **Culture-positive only** — culture-negative keratitis (common, earlier, milder) is unseen.
- **Prevalence.** Cohorts are 51–64% fungal; clinic is fungal-heavy. At real prevalence a fungal call is reliable but a bacterial call is not — a bacterial output is a prompt to confirm.
- **Small test sets** (131 internal, 164 external) → wide CIs; certifying 90% recall $\pm 3\%$ needs ~400 fungal test cases.
- A minor display gap: the extreme-inferior hypopyon meniscus is under-tiled by the $\geq 50\%$ -cornea rule (fixable, low risk).

The ceiling is data, not architecture

Every architectural lever — tile scale, resolution, pooling, lesion-guided tiling, biomarker fusion — has been explored; all but tile-scale and ensembling tied. The consistent conclusion: **~0.81–0.84 is what 682 culture-proven, image-only cases support.**

Next phase — expansion

- **More culture-proven bacterial patients** — the single biggest lever ($\approx 10\times$ a fungal one).
- **Additional organism classes** — moving beyond binary toward the true clinical picture (e.g. viral, Acanthamoeba, mixed), which also addresses the "Others" scope gap.
- **Extend accrual** past 2024, and add Vision-Centre captures (the deployment setting).
- **Intake covariates** (trauma, contact-lens, duration, prior steroid) — information genuinely absent from the pixels, the most plausible route past the image-only ceiling.
- Representation test still open: frozen ViT-B, or fine-tuning last blocks.

Status at this report

A trustworthy, calibrated, interpretable, fungal-safe model with the first honest numbers for the task. The foundation and evaluation harness are in place; the next gains come from data, and that is the work that follows this report.